



Benchmarking road safety performance by grouping local territories: A study in the Netherlands



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ABSTRACT

The method of benchmarking provides an opportunity to learn from better performing territories to improve the effectiveness and efficiency of activities in a particular field of interest. Such a field of interest could be road safety. Road safety benchmarking can include several indicators, ranging from performance indicated by crash statistics, to indicators that also account for consequences in costs or the underlying state of the road safety system and relevant organisation and processes at actor level. The structure and culture of a territory is identified as a basic context of road safety performance. This is regarded as important information to use in grouping of territories to get more homogenous or equal and comparable conditions to learn from 'the best in class'.

The main aim of this study is to assess the usability of different groupings using the physical structure for benchmarking road safety performance at local territorial level. A traditional grouping of municipalities in the Netherlands was compared with a simple grouping of these municipalities based on their level of urbanisation and an advanced grouping in which more indicators such as differences in demography, growth and road structure were taken into account. As in other studies, urbanisation showed to be the most predominant structural factor for grouping local territories and related to differences in road safety performance. However, if information would be needed for specific target groups, other factors like age and gender distribution or the distribution of the road network can provide valuable additional insight and better homogenous starting points for benchmarking. Especially benchmarking of rural territories may profit from such extra distinctive characteristics.

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1. Introduction

Road safety is an important health issue. More than one million people die each year on the world's roads, and between 20 and 50 million sustain non-fatal injuries (WHO, 2013). In May 2011, the Decade of Action for Road Safety 2011–2020 was launched in more than 100 countries, with one goal: to prevent five million road traffic deaths globally by 2020 (WHO, 2013). To achieve this goal, many countries have taken measures towards improving road safety, such as increased enforcement, safer road infrastructure, and new legislation.

To decrease the number of traffic crashes in an efficient way, structured knowledge is needed of the influence of the components of the road traffic system on road crashes.

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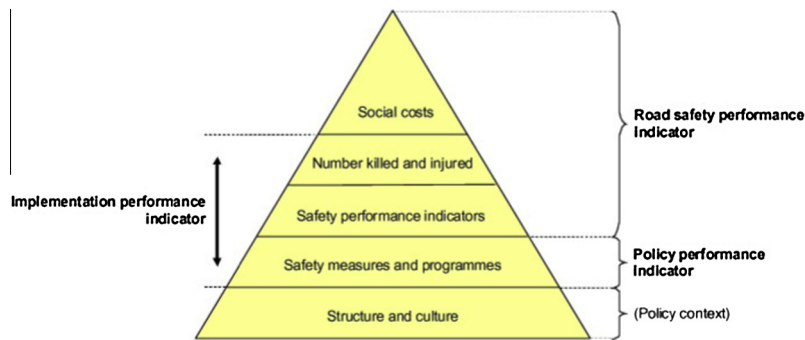


Fig. 1. Road safety pyramid. (LTSA, 2000; Koornstra et al., 2002).

Fig. 1 shows a model of the hypothesised relationships between the components of the road safety system and road crashes, which is visualised by layers of a road safety pyramid (Koornstra et al., 2002; LTSA, 2000). Each layer can be regarded as a result of its lower layer(s). This safety pyramid can be used to understand how road safety performance can be improved. One way to do this is by benchmarking territories (e.g. Bax et al., 2012; Eksler, 2010; Hermans et al., 2009; Wegman et al., 2008; Wegman and Oppe, 2010).

The basic idea behind benchmarking is that it provides an opportunity to learn from 'the best in class'. Wegman and Oppe (2010) state that benchmarking can be done by using indicators at different levels of the road safety pyramid: (a) road safety performance (outcome), (b) implementation performance (a process indicator) and (c) policy performance. In all these performance indicators, the structure and culture layer has to be regarded as a background layer. Fair benchmarking of territories is better possible if these background characteristics are taken into account. Wegman and Oppe (2010) state that "differences in structure and culture between countries not only have an effect on the size and nature of road safety problems, but they also influence the possibilities to reduce the problem effectively and efficiently" (p. 1208). Background characteristics can be taken into account by:

- normalising performance in each territory (e.g. correcting performance by exposure measures), or by
- grouping territories that are comparable on background factors and thus can form a more or less homogenous benchmarking group.

Wegman and Oppe (2010) are in favour of grouping, as territories with similar backgrounds would be more motivated to learn from each other. They studied how European countries could be grouped, using three methods: (a) expert grouping, (b) grouping of road safety level and improvement in road safety using a feature extraction method and (c) grouping based on multiple characteristics of which data was available: seatbelt use, mountainous areas, winter and summer temperature, part of the population with at least secondary education, latitude, main religion, literacy, population density and, gross domestic product. Latitude and population density showed to be important grouping features. Furthermore, the researchers concluded that there were 'many points of agreement' between the different groupings and that grouping did not cause large problems. They however also concluded that a careful selection of variables is needed.

As in the study of Wegman and Oppe (2010), traditionally, comparisons of road safety are made at national territorial level (Eksler, 2010; Jacobs et al., 2000). According to Eksler (2010) and Eksler and Lassarre (2008), benchmarking at sub-national level can provide interesting new opportunities to improve road safety performance. In a previous study, Eksler (2009) showed that 60% of the variance in road mortality between territorial areas was related to differences at the local level.

Eksler (2010), however, also mentions a number of issues have to be taken into account when comparing road safety performance at sub-national level. First of all, because of the low numbers in road deaths, killed and seriously injured (KSI) should be used to get significant results. Secondly, he advises to use relative rates, as rates may largely depend on the constellation within a territory. In his studies directed at sub-national comparisons (see Eksler, 2010; Eksler and Lassarre, 2008; Eksler et al., 2008), he uses Bayesian statistics, also taking spatial and temporal interdependencies between adjacent territories into account. Population density or urbanisation has been found to be of major importance in relation to territorial differences, but from Eksler's studies appears that also other factors are relevant (e.g. Eksler, 2009 and Eksler et al., 2008).

1.1. Structure and culture indicators relevant for road safety

The influence of structure and culture on differences in road safety performance of territories can be described either by an effect on safety risk or by an effect on the amount of exposure to risks. Eksler (2009) classifies three types of structure and culture factors: 'physical structure', 'operational structure', and 'culture'. 'Culture' is related to attitudes, values and norms within a territory, whereas 'operational structures' are aimed at the organisation or the arrangements between potential actors in road safety policy. 'Physical structure' factors can either be static or dynamic. This means that the outcome of

an indicator varies in size through time or not. The following structure and culture factors were identified from literature as relevant for road safety performance: climate, relief, road network, population density, demography, economic situation, modal split, and land use.

1.1.1. Climate

Climate is mentioned by [Wegman et al. \(2008\)](#) as one of the 'structure and cultural factors' that could have an effect on road safety. Weather conditions such as rain and snow are known to have an impact on traffic safety both through an increase of crash risk and through an effect on exposition. During rainy periods the number of bicyclists decreases which results in less bicycle accidents. On the other hand the road surface will have less friction and the line of sight of road users decreases which causes higher risks. Furthermore, [Eksler \(2009\)](#) stated that climate could have an effect on the modal split, e.g. relatively more people will travel by bike at sunny days compared to rainy days. Although climatic differences can be observed between different countries, it is more difficult to see them on lower levels of detail such as between different regions within a country (see also [Eksler, 2010](#); [Wegman et al., 2008](#)).

1.1.2. Relief

A second 'structure and culture factor' that could have an effect on road safety is relief ([Wegman et al., 2008](#)). Relief is a static factor, since it hardly changes over time. Although we did not find a relationship in literature, some features of roads in mountainous areas can be related to road safety. First, the differences in speed between vehicles in mountainous areas can be large, which could result in relatively more overtaking manoeuvres. The curvy roads and the differences in heights between different sections of the road make overtaking manoeuvres more risky since the line of sight is generally smaller than flat roads surfaces. Furthermore, road shoulders can be narrower and steeper in areas with much relief. This increases the probability of a crash. [Eksler \(2009\)](#) also indicated that relief has an influence on traffic safety through its potential effect on the modal split. In general people tend to use motorised vehicles relatively more often in mountainous areas than non-motorised vehicles.

1.1.3. Road network

[Wegman et al. \(2008\)](#) also mentioned road network configurations as a confounding factor. [Dijkstra \(2000\)](#), for instance, found an effect of the structure of residential areas on traffic safety. Other road network variables mentioned in literature that show a strong relationship with road safety performance due to differences in risk or exposition are speed limits (risk), traffic volume (exposition), the ratio between the traffic volume and road capacity (risk and exposition), the density and number of crossings (risk) and road length (exposition) ([Hadayeghi et al., 2003, 2007](#); [Ladron de Guevara et al., 2004](#); [Quddus, 2008](#)). Road network have changed over time. In the Netherlands, the total number of road length did not change very much anymore during the past 10 years, but the number of kilometres per type of road did. Therefore, road network is regarded as a dynamic factor (see also [Eksler, 2009](#)).

1.1.4. Population density

Population density is used in a number of studies as a measure of exposition related to differences in road safety performance between geographic areas. Several studies on predicting crashes found population density to be a significant variable (or even the most important variable) for predicting crashes ([Hadayeghi et al., 2007](#); [Ladron de Guevara et al., 2004](#)). [Eksler et al. \(2008\)](#) used population density as an indication for exposure of traffic to different road types, road network structures and urbanisation and also in a later study, [Eksler \(2009\)](#) found a very strong association between population density and the road fatality rates. Furthermore, population density was used in accident causation models by using the number of households in an area ([Hadayeghi et al., 2007](#); [Ladron de Guevara et al., 2004](#)). Population density is in general a dynamic indicator (see also [Eksler, 2009](#)).

1.1.5. Demography

[Eksler \(2009\)](#) and [Eksler et al. \(2008\)](#) mentions age and gender as potential factors related to differences in road safety performance between regions. The rationale behind this is that the risk of being involved in a traffic crash varies per age and gender, (e.g. in [Gaudry and de Laparent, 2013](#)). [Ladron de Guevara et al. \(2004\)](#) found in a crash prediction model that higher shares of persons under 18 years were related to lower numbers of road crashes. Higher crash risks are reported for male drivers than for female drivers, as well as higher crash rates for drivers aged 16–19 and for drivers aged 75 and older ([DaCoTA, 2012a, 2012b](#); [Massie et al., 1995](#)). Age distribution may vary by geographical region. In general, a higher share of young professionals can be expected in urban areas, whereas in rural areas the average age of inhabitants tends to be older ([Larson et al., 2003](#)). Demography can be seen as a dynamic indicator (see also [Eksler, 2009](#)), although differences in demography over time will probably be relatively small.

1.1.6. Economic situation

The economic situation in an area could influence road safety in various ways. An area with high economic prospects would in general attract more people, stores, and factories, and therefore, generate more traffic. [Eksler \(2009\)](#) used the unemployment rate and [Wiklund et al. \(2012\)](#) used the number of unemployed people as indicator for the economic situation, whereas other authors used factors that described the size of the commercial and industrial sector, such as the number

of employees (Hadayeghi et al., 2007; Ladron de Guevara et al., 2004). The economic situation of an area can be seen as a variable related to differences in traffic density. Furthermore, a poorer region may invest less money in traffic safety measures and in maintenance of roads, which can lead to higher risks. Information on the economic situation vary over time. Therefore, this indicator is considered to be dynamic.

1.1.7. Modal split

The modal split of the vehicle fleet is of interest since some modes of transport such as motorised two wheelers, have higher risks than car drivers (Van Elslande and Elvik, 2012). Travel behaviour can vary between areas because of differences in land use and the road network. On a country level, differences may also arise because of differences in prices of public transport and taxation of fuel and vehicles (Pucher, 1988). Modal split can also be seen as a dynamic factor since other factors, such as the economic situation could influence the modal split, through pricing of fuel, vehicles and maintenance of vehicles.

1.1.8. Land use

Delorme and Lassarre (2005) noted that three major risk factors (speed, drink-driving, belt-wearing) together with urbanisation were related to 80–90% of the gap in road risk levels between France and Great Britain. Eksler (2009) also found a strong association between the share of artificial surfaces with road fatality rates. A plausible cause is that urbanised areas attract more traffic than rural areas and this create more exposure. However, other studies found higher fatality rates in motor vehicle crashes that occurred in less urbanised areas than in more urbanised areas (Baker et al., 1987; Ryb et al., 2012). As relatively more vulnerable road users may be expected on roads in urban areas since distances to commercial activities are in general smaller, they may be overrepresented in urban area crash rates when taking total numbers into account. Although land use may only change slowly, it can be regarded as dynamic (see also urbanisation level in Eksler, 2009).

The usability of these eight 'structure and culture factors' for comparing areas depends on the variance of these factors between the areas under study. For example: the variation of climate and relief might not be large enough to account for differences between areas if comparisons are made between areas that are very similar; however, they might be interesting when comparing the road safety performance between areas that differ a lot on these factors.

1.2. The current study

Road safety management has been decentralized in a number of countries in the last decades. The rationale behind this process is that local authorities generally have a better understanding of the local situation and context to fit in road safety measures. The shift from centralized to a more decentralized approach in road safety policy increases the demand for understanding differences in road safety performance on a regional or even local level.

The main objective of this study is to get insight in whether homogenous grouping of municipalities could be formed as better means to find and benchmark differences in road safety characteristics than traditional grouping based on jurisdictional boundaries of regions (provinces). To study this, physical characteristics of the 'structure and culture level' of the road safety pyramid that are relevant are used for grouping. This was tested with data of municipalities in the Netherlands.

Furthermore, it was tested, whether crash types – a few examples were tested – showed more clear differences between groups in the more homogeneous groupings than in the more traditional grouping of existing territorial jurisdictions. This was tested by comparing the following crash measures per grouping of municipalities:

- (1) the number of serious injury crashes (10 year period) per number of inhabitants;
- (2) crashes that were considered to be more likely in urban areas (i.e. pedestrian crashes, parking crashes and side impact crashes);
- (3) casualties with youngsters (18–24 years of age).

These crash types were chosen to find out whether more homogenous groups of municipalities showed more clear differences on these crash characteristics than municipalities that were grouped in a less homogenous way.

2. Method

2.1. Data

Table 1 provides an overview of the 'structure and culture factors' that have been found in literature to be relevant for road safety. These factors were converted into indicators that are potentially relevant for designing groups of homogenous municipalities for road safety benchmarking in the Netherlands.

Data for the mentioned indicators were gathered from Dutch nationally available sources. The main source of these data was the Dutch Central Bureau of Statistics (CBS). The information on road infrastructure was available from the Dutch National Road Database (NWB). Data on crashes (number of crashes with killed or seriously injured people and the nature

Table 1

List of indicators of structural factors relevant for road safety and the type of data included in this study.

Latent factor	Static/dynamic	Literature	Data in current study
Road network	D	Structure of residential areas: number of crossings with four arms (example)	• Road length per road authority and speed limit. The most important are: share of 30 km/h, 50 km/h, 60 km/h, municipal 80 km/h, provincial 80 km/h and national roads; 2008 information only • Junctions per road length (2008 information and development; Δ2008–1999)
		Speed limits Traffic volume Road capacity-traffic volume ratio Density of crossings Road length	
Population density	D	Household density	• Inhabitants per acre (2008 information and development; Δ2008–1999) • Total population (2008 information and development, Δ2008–1999)
Demography	D	Adres density Population under 18	• Adolescent population: share of people between 18 and 24 years of age (2008 information and development; Δ2008–1999) • Elderly population: share of people aged 75 or higher (2008 information and development; Δ2008–1999)
		Gender differences (not specified) Elderly	
Economic situation	D	Unemployment rate	• Proxy for economic situation: mean economic value of property (2008 information only)
		Number of employees	
Modal split	D	Motorised two-wheelers (example)	• Number of motorised two wheelers per number of inhabitants (2008 information only) • Number of goods vehicles per number of inhabitants (2008 information only)
Land use	D	Urban versus rural	• Share of residential area (2008), • Share of urban facility area (2008) • Share of industrial land use (2008) • Share of recreational land use (2008) • Share of agraric land use (2008) • Share of nature area (2008)

of the crash as registered by the police) was available from the Dutch crash statistics (BRON). Crash data were used on municipality level added up over 10 years (period: 1999–2009) in order to deal with fluctuations over years. This was important to do, as crashes on municipality level fluctuate largely over years.

All records were matched on municipality level. As the boundaries of municipalities could change over years, for instance due to annexation, the situation (borders) in 2011 was taken as the boundary for all data.

2.2. Data analysis

Data analysis was performed with IBM SPSS Statistics version 20, in three steps:

Step 1: analysis of latent factors.

Latent factors in the data (Table 1) were analysed by means of factor analyses, using the principal components extraction method. As the latent factors were considered to be interdependent, factor scores were rotated using promax with Kaiser normalisation (e.g. Gorsuch, 1983). The following criteria were taken into account to decide on the amount and content of factors (e.g. Zwick and Velicer, 1986):

- Factors with an Eigenvalue > 1 (Kaiser, 1960).
- The number of factors that were found before the sharp bend in the scree plot (scree-test criterion; Cattell, 1966).
- Interpretability of the factors.

Several factor scores per municipality were saved. That is: factor scores were saved for factor analyses using several subsets of data. Also factor scores were saved for solutions using the Eigenvalue criterion and the scree-test criterion as long as they provided interpretable latent factors. The factor scores provided new, condensed variables that were used as input for cluster analysis in order to cluster homogeneous or resembling municipalities.

Step 2: clustering of municipalities.

Next, municipalities were clustered into homogenous groups, using hierarchical cluster analysis with Ward's method (Ward, 1963), squared Euclidean distance as a measure for the interval between groups and after having transformed all factor scores per variable in a score between –1 and 1. Hierarchical cluster analysis was performed per set of factor scores in

order to determine the number of municipality clusters that could be distinguished. To reach a final number of groups per analysis, the following criteria were used:

- The number of groups that was found after a relative large change in agglomeration coefficient in the grouping process (e.g. Hair et al., 1998).
- A number of groups larger than two in order to have not too trivial findings of just a group with a few cities and a group with large amount of rural municipalities.
- A number of groups that was not too large in order not to have too complicated distinctions between groups.
- Clusters with at least two municipalities in order to be useful for benchmarking.

When a convenient number of clusters (K) was found per analysis, the final clustering was got via a K -means clustering. Step 3: testing for difference in crash characteristics.

Finally, for the resulting groups of municipalities, it was tested with MANOVA whether the share of crashes per number of inhabitants and the share of urban and rural crashes differed between the municipality groups. The between-group differences were tested using Games-Howell, since unequal variance between groups was assumed. An α of .05 was used.

3. Results

3.1. Grouping

3.1.1. Traditional grouping

The Dutch municipalities are distributed over 12 different provinces, with the number of municipalities per province ranging from $n = 6$ (Flevoland) to $n = 72$ (Zuid-Holland).

3.1.2. Simple grouping

The cluster analysis resulted in several different municipality groupings. In each of these groupings, the latent factor 'urban–rural' was present because it had turned out to be the most important latent factor in all analyses. The factor-scores for this 'urban–rural' factor were summed over all selected clusters per municipality, and municipalities were clustered in K -means analyses with $K = 10$. This resulted in ten different groups of municipalities (n ranging from $n = 4$ in the group with the highest level of urbanisation, to $n = 172$ municipalities in the group with the lowest level of urbanisation).

The urban–rural grouping is indicated with the codes UR1–UR10, where UR1 stands for the most urban group of municipalities, and UR10 for the least urban group.

3.1.3. Advanced grouping

However, some analyses also indicated additional factors to urbanisation level that could be relevant for a clustering of municipalities, such as population characteristics or characteristics of the road network. Furthermore, there was the feeling that the very large group of rural municipalities could be split up further, taking some of these additional factors into account. Therefore, an additional grouping was made by summing the factor scores of the similar latent factors for population characteristics and road network characteristics per K -means analyses per municipality, as was done with the 'Urban–Rural' factor (see Simple grouping).

From this advanced grouping, 276 municipalities were found to form rather homogenous groups. The other 142 municipalities did not clearly match with one of the larger groups and were excluded from the analyses of the additional clusters. Groups containing large cities were allowed to have a small number of members ($n \geq 5$), the groups that contained smaller municipalities had to have a larger amount of members due to their smaller number of crashes. This grouping led to 10 different groups with the numbers of municipalities ranging from $n = 5$ (large cities with relatively high numbers of young inhabitants) to $n = 71$ (rural municipalities with a large share of rural access roads outside the built up area). The following groups were distinguished:

- A-cities: largest cities, large young and growing population, nature and recreational ground use, rural road network with a focus on national through roads and rural access roads with a 80 km/h speed limit.
- B-cities: large cities with somewhat young and growing population, nature and recreational ground use.
- C-cities: cities with somewhat young population, nature and recreational ground use.
- DE-cities: small cities with large young and growing population, nature and recreational ground use.
- Vil ON: villages with older and decreasing population, and a focus on nature as ground use.
- Vil AR60: villages with somewhat younger population with some growth, with a rural road network having a focus on provincial 80 km/h roads and rural access roads with a 60 km/h speed limit.
- R AR80 O: rural municipalities with older and decreasing population, agrarian ground use and a rural road network with a focus on access roads with a 80 km/h speed limit.

- R AR60: rural to very rural municipalities with some agrarian ground use, and a rural road network with a focus on provincial 80 km/h roads and access roads with a 60 km/h speed limit.
- R NTR val: very rural municipalities with valuable estates and a rural road network with a large focus on national trough roads and access roads with a 60 km/h speed limit.
- R AR80: most rural municipalities with agrarian ground use and a rural road network with a focus on provincial 80 km/h roads and rural access roads with a 80 km/h speed limit.

3.2. Comparison of the results between the three grouping types

The effect of the three grouping types was tested by comparing the differences in crash characteristics between the groupings. The next step was to see whether different patterns in crash characteristics appear between different grouping types. As the total number of crashes would be influenced by exposition factors, the first analyses focused on differences in serious crashes adjusted for the number of inhabitants.

3.2.1. Traditional grouping

Municipalities in the province of Zuid-Holland had fewer serious road crashes per number of inhabitants than nearly all other provinces, except for municipalities in the urban provinces of Utrecht and Flevoland (Fig. 2a, Table 2a).

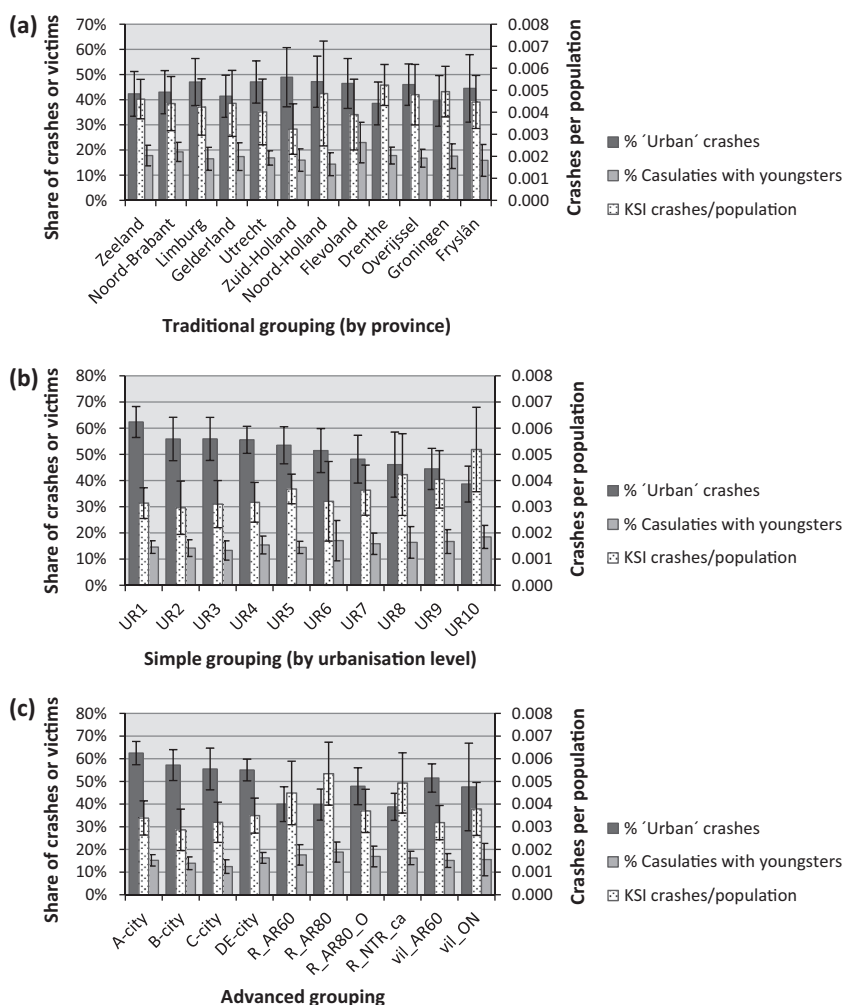


Fig. 2. Serious road crashes (crashes with killed and seriously injured victims (KSI crashes) over a 10-year period) per number of inhabitants, share of 'urban' crashes (pedestrian crashes, parking crashes and side impact crashes), and share of youngsters (18–25 years of age) as victim in a serious road crash for (a) traditional grouping by province ($n = 418$), (b) simple grouping by urbanisation level ($n = 418$) and (c) advanced grouping ($n = 276$). The whiskers show the SD of each group.

Table 2

p-Values of the Games–Howell test of differences in serious crashes per number of inhabitants between (a) the traditional grouping ($n = 418$), (b) simple grouping ($n = 418$) and (c) advanced grouping ($n = 276$). Significant results are marked.

p-Value	Zeeland	L	Gd	U	ZH	NH	Fl	D	O	Gr	Fr
N-Brabant (NB)	1.000										
Limburg (L)	.994	1.000									
Gelderland (Gd)	1.000	1.000	1.000								
Utrecht (U)	.926	.990	1.000	.994							
Z-Holland (ZH)	.005*	.000*	.014*	.000*	.421						
N-Holland (NH)	1.000	.975	.907	.988	.725	.001*					
Flevoland (Fl)	.991	.999	1.000	.999	1.000	.991	.952				
Drenthe (D)	.831	.282	.212	.393	.134	.000*	.998	.739			
Overijssel (O)	1.000	.975	.907	.990	.723	.001*	1.000	.959	.990		
Groningen (Gr)	.997	.740	.602	.857	.405	.000*	1.000	.901	.999	1.000	
Fryslân (Fr)	1.000	1.000	1.000	1.000	.986	.002*	.998	.998	.583	.999	.956

p-Value	UR1	UR2	UR3	UR4	UR5	UR6	UR7	UR8	UR9
UR2	1.000								
UR3	1.000	1.000							
UR4	1.000	1.000	1.000						
UR5	.786	.345	.287	.498					
UR6	1.000	1.000	1.000	1.000	.961				
UR7	.862	.474	.472	.689	1.000	.982			
UR8	.253	.027*	.014*	.034*	.658	.386	.603		
UR9	.334	.025*	.002*	.013*	.669	.481	.621	1.000	
UR10	.027*	.000*	.000*	.000*	.000*	.001*	.000*	.030	.000*

p-Value	A-city	B	C	DE	R60	R80	R800	RN	Vil60
B-city (B)	.954								
C-city (C)	1.000	.991							
DE-city (DE)	1.000	.687	.987						
RAR60 (R60)	.254	.002*	.000*	.007*					
RAR80 (R80)	.028*	.000*	.000*	.000*	.003*				
RAR800 (R800)	.998	.448	.857	1.000	.203	.000*			
RNTRval (RN)	.134	.004*	.011*	.047*	.972	.983	.175		
Vilar60 (vil60)	1.000	.997	1.000	.992	.024*	.000*	.903	.022*	
Vil_ON	.995	.473	.849	.998	.583	.007*	1.000	.353	.887

3.2.2. Simple grouping

In the simple grouping of municipalities, the most rural group showed to have more serious crashes per number of inhabitants than all other groups (Fig. 2b, Table 2b). Also the second and third rural group had a higher crash rate compared to most groups of more urban municipalities.

3.2.3. Advanced grouping

The advanced grouping of municipalities showed that the amount of serious crashes per number of inhabitants was larger in rural municipalities with a focus on 80 km/h rural access roads than in most other groups. (Fig. 2c, Table 2c). Rural municipalities with a focus on 60 km/h access roads and those with a focus on trough roads, showed a larger number of serious crashes per number of inhabitants than most city-groups.

3.3. Share of 'urban' crash types

3.3.1. Traditional grouping

Municipalities ordered by province showed only a few and less clear results (Fig. 2a and Table 3a): municipalities in the provinces of Brabant, Gelderland and Groningen showed to have lower shares of severe urban crashes than municipalities in the Zuid-Holland. The former provinces are more rural and the latter is more urban.

3.3.2. Simple grouping

The most rural municipalities turned out to have the lowest share in registered severe urban crashes (pedestrian, parking, side impact) compared to all other municipality groups (Fig. 2b and Table 3b). Also other rural municipality groups showed to have lower shares of urban crashes than more urban municipality groups.

3.3.3. Advanced grouping

The share of registered severe urban crashes was higher in A-cities compared to all rural municipalities (see Fig. 2c and Table 3c), and higher in all other cities compared to most rural municipalities, except those with a large elderly population.

Table 3

p-Values of the Games–Howell test of differences of serious urban crashes between (a) the traditional grouping (*n* = 418), (b) simple grouping (*n* = 418) and (c) advanced grouping (*n* = 276). Significant results are marked.

<i>p</i> -Value	Zeeland	L	Gd	U	ZH	NH	Fl	D	O	Gr	Fr
N-Brabant (NB)	1.000										
Limburg (L)	.896	.635									
Gelderland (Gd)	1.000	.996	.184								
Utrecht (U)	.895	.637	1.000	.193							
Z-Holland (ZH)	.483	.037*	.999	.002*	.193						
N-Holland (NH)	.839	.379	1.000	.054	.999	.999					
Flevoland (Fl)	.998	.998	1.000	.967	1.000	1.000	1.000				
Drenthe (D)	.992	.858	.216	.993	1.000	.051	.162	.843			
Overijssel (O)	.980	.921	1.000	.482	.217	.964	1.000	1.000	.379		
Groningen (Gr)	.999	.937	.206	1.000	1.000	.024*	.127	.898	1.000	.424	
Fryslân (Fr)	1.000	1.000	.999	.992	.208	.927	.999	1.000	.866	1.000	.934
<i>p</i> -Value	UR1	UR2	UR3	UR4	UR5	UR6	UR7	UR8	UR9		
UR2	.726										
UR3	.669	1.000									
UR4	.586	1.000	1.000								
UR5	.393	.996	.990	.993							
UR6	.231	.863	.764	.758	.999						
UR7	.092	.132	.036*	.021*	.483	.949					
UR8	.049*	.041*	.009*	.006*	.188	.657	.998				
UR9	.053	.002*	.000*	.000*	.007*	.084	.572	.999			
UR10	.027*	.000*	.000*	.000*	.000*	.000*	.000*	.021*	.000*		
<i>p</i> -Value	A-city	B	C	DE	R60	R80	R800	RN	Vil60		
B-city (B)	.767										
C-city (C)	.493	1.000									
DE-city (DE)	.274	.994	1.000								
RAR60 (R60)	.003*	.000*	.000*	.000*							
RAR80 (R80)	.003*	.000*	.000*	.000*	1.000						
RAR800 (R800)	.014*	.096	.299	.147	.056	.047*					
RNTRval (RN)	.001*	.000*	.000*	.000*	1.000	1.000	.051				
VilAR60 (vil60)	.103	.673	.944	.901	.016*	.015*	.966	.009*			
Vil_ON	.272	.764	.907	.906	.888	.875	1.000	.819	.999		

The rural municipalities with a rural 80 km/h road network had a lower share of urban road crashes than similar municipalities with a large elderly population. Except for this last finding, this was largely what was expected.

3.4. Serious youngster casualties

3.4.1. Traditional grouping

The grouping of municipalities per province showed a higher share of casualties among youngsters in the province of Brabant compared to the urban provinces of Noord-Holland and Zuid-Holland (Fig. 2a, Table 4a).

3.4.2. Simple grouping

The simple urban grouping of municipalities showed a larger share in youngsters in the most rural group than four of the more urban municipalities (Fig. 2b, Table 4b). The second most rural group of municipalities showed to have a larger share of casualties among youngsters than the third most urban group of municipalities. This is contrary to what was expected when taking exposure (share in population) into account.

3.4.3. Advanced grouping

In the advanced grouping of municipalities, it turned out that the second and third most urban municipalities (B- and C-cities) had a lower share in casualties among youngsters than the rural municipalities with a focus on rural access roads and not a very old population (Fig. 2c, Table 4c). C-cities also had a lower share than DE-cities.

4. Conclusions and discussion

The main objective of this study was to see whether homogenous grouping of local territories would provide a better basis for benchmarking differences in road safety characteristics than more traditional comparisons based on jurisdictional boundaries of regions (provinces). The road safety pyramid model (Koornstra et al., 2002; LTSA, 2000) was taken as the theoretical framework, and differences were based on physical components from the 'structure and culture level' of this pyramid.

Table 4

p-Values of the Games–Howell test of differences in serious youngster casualties between (a) the traditional grouping ($n = 418$), (b) simple grouping ($n = 418$) and (c) advanced grouping ($n = 276$). Significant results are marked.

p-Value	Zeeland	L	Gd	U	ZH	NH	Fl	D	O	Gr	Fr
N-Brabant (NB)	.983										
Limburg (L)	.998	.148									
Gelderland (Gd)	1.000	.584	1.000								
Utrecht (U)	1.000	.062	1.000	1.000							
Z-Holland (ZH)	.940	.001*	1.000	.931	.993						
N-Holland (NH)	.309	.000*	.591	.079	.118	.661					
Flevoland (Fl)	.903	.978	.732	.838	.764	.649	.445				
Drenthe (D)	1.000	.960	.995	1.000	.999	.871	.171	.896			
Overijssel (O)	1.000	.165	1.000	1.000	1.000	.999	.318	.760	.999		
Groningen (Gr)	1.000	.928	1.000	1.000	1.000	.966	.273	.873	1.000	1.000	
Fryslân (Fr)	.989	.327	1.000	.995	1.000	1.000	.992	.679	.981	1.000	.995

p-Value	UR1	UR2	UR3	UR4	UR5	UR6	UR7	UR8	UR9
UR2	1.000								
UR3	.998	.997							
UR4	1.000	.988	.667						
UR5	1.000	1.000	.960	.995					
UR6	.975	.905	.659	.997	.930				
UR7	.998	.863	.274	1.000	.886	1.000			
UR8	.959	.762	.249	.999	.792	1.000	1.000		
UR9	.804	.253	.012*	.938	.170	1.000	.994	1.000	
UR10	.300	.002*	.000*	.060	.000*	.998	.049*	.558	.0125

p-Value	A-city	B	C	DE	R60	R80	R800	RN	Vil60
B-city (B)	.990								
C-city (C)	.575	.937							
DE-city (DE)	.994	.430	.008*						
RAR60 (R60)	.639	.032*	.000*	.767					
RAR80 (R80)	.258	.003*	.000*	.053	.707				
RAR800 (R800)	.983	.564	.083	1.000	1.000	.865			
RNTRval (RN)	.998	.589	.035*	1.000	.888	.189	1.000		
VilAR60 (vil60)	1.000	.994	.568	.991	.562	.159	.978	.996	
Vil_ON	1.000	.998	.869	1.000	.982	.785	1.000	1.000	1.000

A number of studies have explored the comparison of regions that are similar in one or more relevant factors (e.g. Eksler, 2010; Hermans et al., 2009; Wegman and Oppe, 2010). Most of these studies have explored groupings of regions on a theoretical level; practically, they have mainly been used to group countries. However, some studies have reported benchmarking on sub-national level (Eksler, 2009, 2010; Eksler et al., 2008). These studies were mainly directed at finding explanations for territorial differences in an improvement of road safety over time, and studying the relevance of benchmarking road safety at national and sub-national levels. Regarding benchmarking road safety performance, Wegman and Oppe (2010) state that territories have first to be grouped using 'structure and cultural factors' in order to get more homogenous groups. This would provide better motivation for learning from 'the best in class'.

Three levels of grouping were explored in this study: (1) a traditional grouping of local territories based on the region where they are located in (province level), (2) a simple grouping based on urbanisation level and (3) an advanced grouping in which more variables were taken into account. Differences in road safety performance were analysed, using three road safety indicators as an example:

- (1) the number of KSI crashes per population,
- (2) the share of crashes that were expected to be more likely in more urban regions (i.e. pedestrian crashes, parking crashes and side impact crashes),
- (3) the share of casualties with youngsters (18–24 years of age) who are known to be a high risk group.

Irrespective of these indicators, a more clear pattern of differences in these indicators was expected in the more homogenous groupings than in the traditional, more heterogeneous grouping.

Out of all factors that were included in this study, the level of urbanisation turned out to be the most important one for homogenous grouping of local territories and related to differences in road safety performance. This finding supports the findings of other benchmarking studies (e.g. Delorme and Lassarre, 2005; Eksler, 2009; Eksler et al., 2008), but in which a much smaller set of factors was included. In this study, we found differences between urban and rural regions, both for the number of KSI crashes per population and the share of urban type of crashes. However, regarding crash performance of specific demographic groups (young road users in this study), less clear effects were found: municipalities with large shares of young and risky road users (between 18 and 24 years of age) did show relatively large shares of this group in

serious casualties in the more rural areas where generally a more elderly population is living. An explanation might be that youngsters drive more and/or have higher risks in rural areas than in more urban areas. This could also be related to cultural differences, for instance reflected in driving styles. Data of such factors were not part of this study. Reasons for counterintuitive findings could be explored further.

The results of this study show that grouping local territories into homogenous areas based on urbanisation level can have an added value. Benchmarking territories in one group with similar values for urbanisation could lead to more interesting outcomes than a traditional grouping based on regional boundaries as a starting point to gain more insight in relative performance. In this way, grouping similar areas can act as a way of normalising for exposure, for instance 'urban' exposure.

The added value of the advanced level of grouping seems to be low compared to the urban grouping. The analyses that were performed to find latent factors before grouping the municipalities, already showed that urbanisation level was the most important factor and could be indicated – depending on the selection criteria – to be the only important factor, related to most variance in the available structure and culture data of the municipalities.

A limitation in analyses that deal with available statistics, and therefore also a limitation in studies where areas are grouped, is the limited availability of usable and reliable data. This study has been conducted on the aggregation level of municipalities in the Netherlands. A lot of data is present on this aggregation level, but there is a lack of some data such as the number of vehicle kilometres travelled. Furthermore, any bias in the results due to regional differences in crash recordings by the police cannot be ruled out. Another issue related to data is the increase in scale of municipalities: some cities showed to have gained so much rural territory, that in the share of data, they are not classified as city any more. There is not found a solution yet for areas like municipalities with a scatter of villages within its border versus one main city centre. It is imaginable that these situations, that are expressed in a similar way in the data that was used, are in reality very different, especially in the traffic flows that they attract.

This study focused on structural characteristics within the 'structure and culture level' that could be found as relevant for road safety and available in data. Cultural factors were not taken into account as such. This could have offered additional interesting results, although more clear results could be expected when comparing countries instead of comparing within-country areas because of larger differences in the former. Another issue when taking cultural factors into account, is the operationalization of such factors and data that should be available for a benchmarking study.

The road safety pyramid model contains a number of other levels that are interesting to look at and may be useful for finding explanations for differences in road safety performance between groups of territories. These levels are the level of 'road safety measures' and the level of 'safety performance indicators'. In future benchmarking studies, characteristics on this level are interesting to take into account as well, as was done in a recent benchmarking study of European countries (Bax et al., 2012).

We conclude that grouping local territories (e.g. municipalities) based on physical structure factors may indeed be valuable for benchmarking purposes. If municipalities are not comparable for the most predominant factors that influence road safety, traditional benchmarking based on geographical boundaries (e.g. grouping by provincial boundaries), may not reflect the actual road safety status. First of all, this study and many other studies point out that it would be preferable to compare the road safety situation between groups of comparable territories that have at least similar levels of urbanisation. However, if information would be needed for specific target groups, other factors like age and gender distribution or the distribution of the road network might provide better insight. Benchmarking of rural territories may especially need such extra distinctive characteristics. Particularly with territories that are classified as 'rural', it should be kept in mind that – at least in the Netherlands where this study was performed – municipalities can be heterogeneous within their boundaries, that is: having a very urban centre surrounded by a vast area of rural area (see also Eksler et al., 2008; Yannis et al., 2013). In this study, we did not have data to express differences between rural municipalities in which one large urban centre was located versus rural municipalities with a scatter of smaller urban centres within its boundaries.

The quality of the benchmarking process depends not only on the quality and availability of the data, but also on the decision of how to specify different groups of similar study areas and the size of the groups. For these two issues, interpretation by researchers is still needed, and several outcomes are possible. All these issues should be kept in mind when conducting or interpreting the results of benchmark studies.

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